

CHAPTER 9

Exercises

All the questions from the document in one place.

Chapter 1

Exercise 1.1.1

To get used to natural units, let's try some examples. The mass of the electron is 0.511 MeV. What is its mass in kg? The same electron has a momentum of 1 MeV. What are its energy and momentum in SI units?

Exercise 1.1.2

Using dimensional analysis arguments and the natural units defined above, show that the natural units for time and length are eV^{-1} .

Exercise 1.2.1

Given an electron microscope uses electrons with energies 10^6 larger than that of visible light, what length scale would it be able to resolve?

Exercise 1.2.2

What is the uncertainty on the momentum of an electron inside an atom? Is it relativistic?

Exercise 1.2.3

What is the uncertainty on the momentum of a quark inside a proton? Is it relativistic? The quarks have masses of a few MeV. What can we conclude about the origins of the mass of the proton?

Exercise 1.3.1

At the large hadron collider, protons are collided together. What is the gravitational force between two protons separated by the diameter of a proton? How does this compare to the coulomb repulsion between them? Just use orders of magnitude to compare.

Exercise 1.3.2

To demonstrate the uncertainty principle in Eq.[1.12], consider a propagator moving at c . If the mass of the propagator is around 100 GeV, what order of magnitude would the range be? Use this to comment on the range of the weak force.

Exercise 1.3.3

A good model for the nuclear force is the exchange of pions (π) between nucleons. Estimate the range of the nuclear force.

Exercise 1.3.4

The momentum of a typical football in a professional-level penalty shoot out is about 2.5×10^{19} GeV. What is its velocity in m s^{-1} if it weighs $2.011 \times 10^{-31} M_{\odot}$ ($1 M_{\odot} = 1 \text{ solar mass} = 1.989 \times 10^{30} \text{ kg}$)?^a

^aHow you are feeling now is how I feel when I see an answer in joules to a particle physics question asking for the energy... ☺

Chapter 2

Exercise 2.2.1

Draw the Feynman diagram for two electrons scattering off each other.

Exercise 2.2.2

This should be attempted after we have covered relativistic kinematics. Show that the process $e \rightarrow e\gamma$ doesn't conserve both momentum and energy.

Chapter 3**Exercise 3.1.1**

Starting with the definition of proper velocity, derive Eq. [3.5].

Exercise 3.1.2

Show that Equation [3.8] yields the classical result in the limit at $v \ll c$.

Exercise 3.2.1

In a reference frame Σ , a particle has a momentum of $\vec{p} = [0, 50 \text{ GeV}/c, 0]$ and a mass of 27 GeV. In a reference frame Σ' moving relative to Σ with velocity $\vec{v} = [0.99c, 0, 0]$, what would the momentum be measured as? What about in the frame Σ'' moving relative to Σ with velocity $\vec{v}_2 = [0, 0.99c, 0]$? What about in the frame Σ''' moving relative to Σ with velocity $\vec{v}_3 = [0, 0.88c, 0]$?

Exercise 3.4.1

The large hadron collider collides protons each with a momentum of 7 TeV. What is the centre of mass energy $\sqrt{s}$?

Exercise 3.4.2

In reality, it is the quarks which scatter at the LHC, rather than the protons themselves. What is $\sqrt{s}$ if instead one of the interacting quarks carries 10% of its proton's momentum, and the other carries 30% of the other proton's momentum?

Exercise 3.4.3

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What proton momentum would be needed to obtain the same centre of mass energy if the second proton was instead at rest? We call this a fixed target experiment.

Exercise 3.5.1

If the muons can make it, they make it in all frames, but from the muon's frame of reference it's life time is not dilated. How do they make it?

Exercise 3.6.1

Can a particle decay into only one particle?

Exercise 3.7.1

A fictional particle with a mass of 42 GeV, decays to a pair of particles which each have a mass of 6 GeV. What are the momenta of the produced particles?

Exercise 3.7.2

I throw together two masses, each with a mass m , and they collide head-on. Each had an initial velocity of $0.4c$. What is the mass of the combined object after the collision?

Exercise 3.7.3

A particle at rest with mass m_1 decays into two particles each with mass m_2 . What is the speed of the decay products? What condition must hold for this to occur?

Exercise 3.7.4

A particle at rest of mass m_1 decays into two particles with masses m_2 and $m_3 = 0$, respectively. What is the speed of the particle with mass m_2 ?

Chapter 4

Exercise 4.2.1

The isotope carbon-11 (^{11}C) decays emitting a positron and is used in positron emission tomography (PET) imaging. This works by detecting the photons produced by the annihilation of the positron with an electron, $e^+ + e^- \rightarrow \gamma\gamma$. Calculate the energies of the two photons. State any assumptions made.

Exercise 4.2.2

Draw the lowest order Feynman diagram for the process $e^+ + e^- \rightarrow \gamma\gamma$.

Exercise 4.3.1

Draw the Feynman diagram for the reaction $p\bar{p} \rightarrow n\bar{n}$, where p and n are protons and neutrons. Remember to include in your diagram the spectator quarks.

Exercise 4.4.1

Given the mass of the W boson is approximately 80 GeV, what is the approximate range of the W -mediated weak force?

Exercise 4.4.2

Given the masses of the quarks and leptons (see earlier in the notes), write down all kinematically allowed the decay modes of the W^- boson, which has a mass of 80.4 GeV. Remember that charge must be conserved.

Exercise 4.7.1

Draw the Feynman diagrams for the processes $e^+ e^- \rightarrow \mu^- \mu^+$ when it is γ and Z mediated and derive the strength of the processes. You may assume that the invariant mass of the mediator's energy-momentum vector is much less than the mass of the Z .

Chapter 5

Exercise 5.4.1

Consider the below reactions and decide which are allowed and which are not based on conservation rules. If they are allowed by conservation rules draw a Feynman diagram for the dominant process.

If you are unsure of any particles details, use the notes or the Particle Data Group pages to find them https://pdg.lbl.gov/2025/listings/contents_listings.html.

1. $n \rightarrow pe^-\bar{\nu}_\mu$
2. $\mu^+ + \mu^- \rightarrow \gamma\gamma$
3. $D^+ \rightarrow K^-\pi^-\pi^+$
4. $K^0 \rightarrow \pi^-e^+\nu_e$
5. $pp \rightarrow \Delta^0 p\bar{p}\pi^+$
6. $\Delta^0 \rightarrow p\pi^-$
7. $\tau^- \rightarrow \mu^-\bar{\nu}_\mu$
8. $\rho^0 \rightarrow \pi^+\pi^-$

Chapter 6

Exercise 6.3.1

Using conservation rules, show that the π^+ can only decay through the weak force, whereas the π^0 can also decay electromagnetically. Can pions decay via the strong force?

Exercise 6.4.1

Show that the D^{*+} can undergo this decay. Check both the conservation laws and kinematics. The D^{*+} is an excited D meson with a mass of 2010 MeV, use the PDG to get any other information you need https://pdg.lbl.gov/2025/listings/contents_listings.html.

Exercise 6.6.1

Draw the lowest order Feynman diagram for the decay of the Ξ^- .

Exercise 6.4.2

Draw the Feynman diagram for the D^{*+} decay to $D^+ + \pi^0$, and show this also satisfies conservation rules and kinematics.

Exercise 6.6.2

Draw the lowest order Feynman diagram for the decay of the Ξ^0 .

Exercise 6.6.3

Draw the lowest order Feynman diagrams for the decays of the Ω^- shown above.

Chapter 7

Exercise 7.5.1

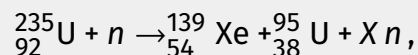
Calculate the mass deficit and binding energy of ${}^{62}_{28}\text{Ni}$. 61.928345 u.

Exercise 7.5.2

Calculate the mass deficit and binding energy of ${}^{60}_{27}\text{Co}$. $M = 59.933817$ u.

Exercise 7.8.1

In the fission reaction



find the number of neutrons released X . Calculate the total energy released in the reaction. How does neutron absorption help trigger this reaction? Hint: consider how energy changes when a neutron is added.

Chapter 8

Exercise 8.6.1

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Consider $^{57}_{27}\text{Co}$, which is unstable. Is this an odd-odd, odd-even, or even-even nucleus? Can it undergo β decay and/or electron capture? Show your workings. The mass of $^{57}_{27}\text{Co}$ is 56.936291 u and for $^{57}_{26}\text{Fe}$ is 56.935394 u.

Exercise 8.6.2

Show that the nuclide ^{60}Co is unstable with respect to β decay. The mass of ^{60}Co is 59.933817 u and of ^{60}Ni is 59.930786 u.